NITROGENASE

Nitrogenase is an enzyme that helps convert nitrogen gas into ammonia.
The function of nitrogenase is to convert atmospheric nitrogen gas (N_2) into ammonia (NH_3), making nitrogen usable for living organisms.

Nitrogen gas (N_2) is not usable because it is very stable and hard to break apart.

Why?
The two nitrogen atoms in N_2 are held together by a very strong triple bond
Most organisms don't have enough energy or the right enzymes to break this bond.

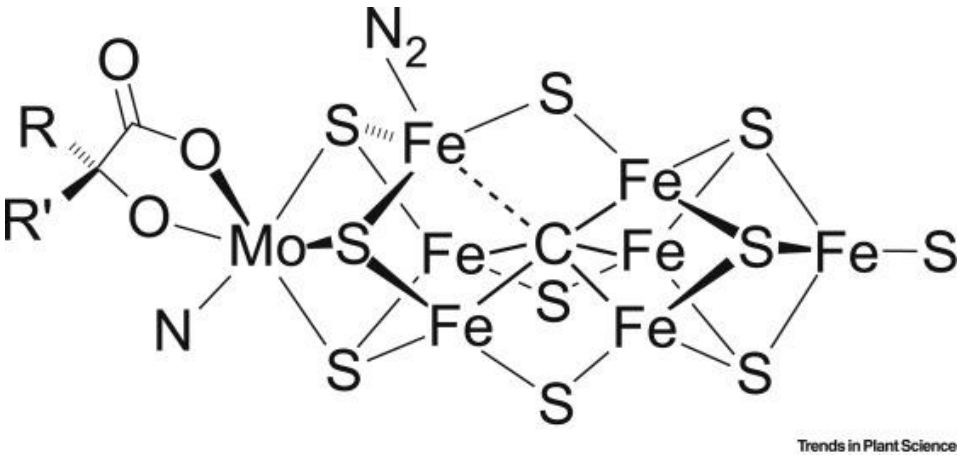
- Nitrogenase is special—it can break that strong bond
- It changes N_2 into ammonia (NH_3), which is easier for organisms to use

Nitrogenase uses energy and electrons to break nitrogen gas and convert it into ammonia.

Nitrogenases are enzymes that are produced by certain bacteria, such as **cyanobacteria** (blue-green bacteria) and **rhizobacteria**. These enzymes are responsible for the reduction of nitrogen (N2) to ammonia (NH3). Nitrogenases are the only family of enzymes known to catalyze this reaction, which is a step in the process of **nitrogen fixation**. Nitrogen fixation is required for all forms of life, with nitrogen being essential for the biosynthesis of molecules (nucleotides, amino acids) that create plants, animals and other organisms. They are encoded by the **Nif genes** or **homologs**. They are related to **protochlorophyllide reductase.**

Nitrogenase converts nitrogen gas (N_2) into ammonia (NH_3) through the process of nitrogen fixation. First, nitrogen gas binds to the nitrogenase enzyme found in certain bacteria such as Rhizobium. The enzyme then uses energy from ATP and transfers electrons to the nitrogen molecule. This energy and electron supply allow nitrogenase to break the very strong triple bond (N≡N) in nitrogen gas. After breaking the bond, hydrogen atoms are gradually added to the nitrogen atoms through a series of reactions until ammonia is formed. Finally, the ammonia is released and can be used by plants to produce proteins and other essential compounds.

How to explain this nitrogenase structure (script for PPT)



“This image shows the structure of nitrogenase, the enzyme responsible for nitrogen fixation.”

“Inside nitrogenase, there is a special region called the active site, which contains metals like iron (Fe) and molybdenum (Mo). This part is very important because it is where nitrogen gas is converted into ammonia.”

“Surrounding the active site are sulfur (S) and iron clusters, which help transfer electrons during the reaction.”

“The nitrogen molecule (N_2) binds to this active site and is gradually broken apart. Because of this structure, nitrogenase can perform one of the most difficult reactions in nature — breaking the strong bond in nitrogen gas.”

What is the double bond (= O)?
The double bond to oxygen (C=O) is called a carbonyl group.
It means a carbon atom is strongly bonded to oxygen with two shared electron pairs.